# Sandipan Seal

**AI / Machine Learning Engineer — LLM & Agentic Systems**

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## Summary

AI Engineer at aiio GmbH with 4 years building **production LLM and agentic systems** (RAG, multi-agent, knowledge graphs), delivered directly to clients as a **forward-deployed engineer**. Master's research on explainable, LLM-based RF-jamming analysis.

## Skills

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| **Languages** | Python, TypeScript, JavaScript, SQL, Go, C++ |
| **Generative AI** | RAG, multi-agent & agentic systems, prompt engineering, function calling, fine-tuning, knowledge graphs, MCP, Claude Agent SDK |
| **Evaluation & Safety** | LLM-as-judge, RAGAS, guardrails (PII redaction, prompt-injection defense), LLMOps, GDPR |
| **LLM Frameworks** | LangChain, CrewAI, Hugging Face, sentence-transformers, Jina embeddings, OpenAI, Claude, Ollama, vLLM, n8n |
| **Machine Learning** | PyTorch, TensorFlow, scikit-learn, pandas, NumPy, NLP, computer vision, explainable AI |
| **Backend & Frontend** | FastAPI, Node.js, REST APIs, GraphQL, React, Next.js, Tailwind CSS |
| **Databases** | PostgreSQL, MongoDB, Redis, Qdrant, FalkorDB (Cypher), Neo4j, Supabase |
| **Cloud & DevOps** | AWS (Lambda, API Gateway, S3, SQS, Glue, Athena), Terraform, Docker, Kubernetes, GitHub Actions, Render, Vercel |
| **Monitoring & Testing** | OpenTelemetry, Prometheus, Grafana, Langfuse, Playwright, Selenium |

## Experience

**AI Engineer — aiio GmbH, Magdeburg, Germany** *Oct 2026 – Present*

- Gave clients **70–80% more clean, usable data** by building and deploying end-to-end **RAG-enabled multi-agent systems** that ingest process data, extract workflows, and use LLMs to build **knowledge graphs** and chatbots.
- Cut clients' consulting spend by researching and shipping software products that act as an **automated consultant**, so they no longer hire multiple consultants.
- Turned ambiguous client problems into working solutions as a **forward-deployed engineer**: gathered requirements directly with clients, then designed, built, and deployed the full system.

**Working Student, AI/ML Software Development — aiio GmbH** *Jan 2025 – Sep 2026*

- Added **AI suggestions, recommendations, and AI-generated descriptions** to aiio's main process-data product and Polaris by building LLM-based features into the live products.
- Accelerated delivery of customer-facing AI tools by building responsive **React.js / Next.js** front-ends integrated with production models.

**Senior Software Engineer — Persistent Systems Ltd, Kolkata, India** *Sep 2021 – Mar 2024*

- Raised healthcare-diagnostic prediction accuracy by ∼20% for client Adaptive Biotechnologies by designing and deploying **predictive ML models** into production.
- Delivered enterprise features end to end for PerkinElmer by owning **React, .NET/C#, and MongoDB** work from API design to production deployment.
- Cut regression-test cycle time by ∼30% by building automated **QA suites** (Selenium, API testing); twice awarded the **High Five Team Award**.

## Selected Projects

**Master's Thesis, OVGU Magdeburg**

*From RF Embeddings to Semantic Reasoning: Explainable Jamming Analysis via LLM-Based Evidence Interpretation*

- Raised LLM jamming-regime accuracy from chance (19%) to **56%** (**73%** with a zero-cost deterministic correction rule; best of five LLMs 78%) on a class-balanced 2,500-window test set by replacing salience-ranked embedding evidence with a supervised **four-axis LDA representation**.
- Showed that **100%-grounded citations were not always causally faithful** by designing paired perturbation tests (masking, necessity, sufficiency, counterfactuals, matched random controls) with McNemar/Holm and group-aware bootstrap statistics across five LLMs.
- Established the LLM's real contribution by benchmarking it against conventional ML on identical windows: the same four LDA features without an LLM reach 80–81% (vs. 74% for 768-feature logistic regression), so the LLM adds per-decision explanation, not accuracy.

**InferOps-AI — Production LLM Gateway** GitHub

- Cut LLM inference cost and blocked unsafe requests by building a gateway with **cost-aware model routing, RAG** (Qdrant/pgvector), **PII redaction**, and **prompt-injection guardrails**, verified by a deterministic +

LLM-as-judge + **RAGAS** eval suite at **>80% test coverage**.
- Shipped to production on **AWS serverless** (Lambda, API Gateway, CloudFront, S3) via **Terraform**, with **Langfuse** tracing and Prometheus/Grafana observability.

**AegisOps-AI — Agentic Incident Commander** GitHub

- Reduced incident-triage effort by building a **multi-agent** system for automated **root-cause analysis** over logs, metrics, and **Kubernetes** state, with confidence scoring and approval-gated runbooks (**human-in-the-loop**).
- Made incident impact visible by adding SLA tracking, a lifecycle state machine, and service-dependency / blast-radius analysis (FastAPI, Next.js, OpenTelemetry).

**DeliveryLens-AI — Delivery Intelligence Platform** GitHub

- Surfaced engineering workflow bottlenecks and measured intervention ROI by connecting **Jira and GitLab** to an AWS serverless data lake (Glue, Athena, Step Functions) with **guarded Text2SQL** analytics (DSPy) and human-in-the-loop safeguards.

**AI Automation Testing Tool — Multi-Agent QA (CrewAI)** GitHub

- Cut manual test creation by building a **CrewAI multi-agent** system (explorer, scenario writer, script generator, auto-repairing runner) that turns natural-language requirements into executable **Playwright** tests, with live SSE progress streaming.

**Additional projects:** AI Compliance Gatekeeper (**n8n** + FastAPI — PII/policy classification & **data-leakage prevention**) · Mine@TENJI (German parliamentary speech recommender — **SentenceTransformers**, **FAISS**). All on GitHub.

## Education

**M.Sc. Digital Engineering (Computer Science), OVGU Magdeburg** *Apr 2024 – Oct 2026 (expected)*
**B.Tech. Electronics & Communication Eng., Narula Institute of Technology** *Aug 2017 – Jun 2021*

## Awards & Patent

**Patent:** Low-Cost Automated Bag-Valve-Mask Ventilation Machine — Indian Patent IN 202031037602 A (2020)
**Awards:** Conquer COVID-19 Hackathon Winner (2020) · Best Paper ×2 (I3SET 2019, NCMR 2020) · High Five Team Award ×2 (Persistent)

## Certifications

Claude Code in Action — Anthropic (2026) · Fine-tuning LLMs — Hugging Face (2026) · AI Agent Fundamentals — Databricks (2026) · Certified Ethical Hacker (CEH) — EC-Council (2020)

## Languages

Bengali (Native, C2) · English (IELTS, C1) · German (Conversational)